

Logistic regression

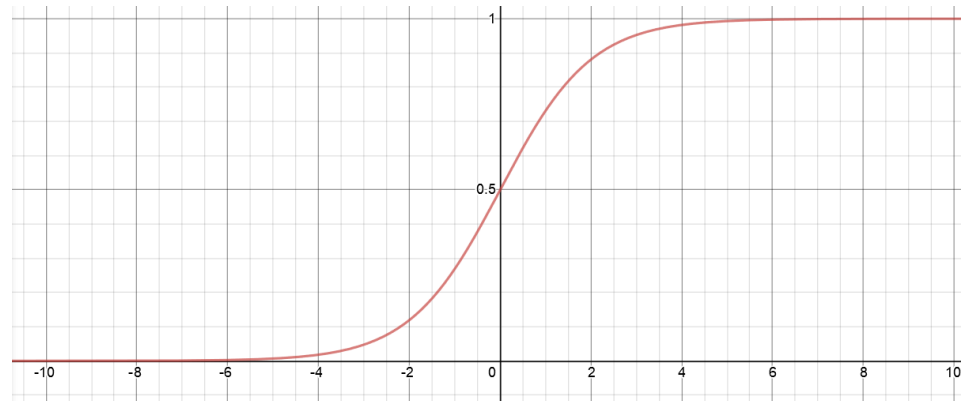
Lee-Ad Gottlieb

Learning

- Classification: $\{0,1\}$
 - Example: SVM gives $\text{sign}(wx)$
- Regression: $[0,1]$
 - Example: Linear regression gives wx
- Maximum Likelihood: Probability that 1 is true
 - (usually done for two classes $\{0,1\}$)
 - Example: **Logistic regression**
 - Can do classification too: $[\text{Pr} > .5]$

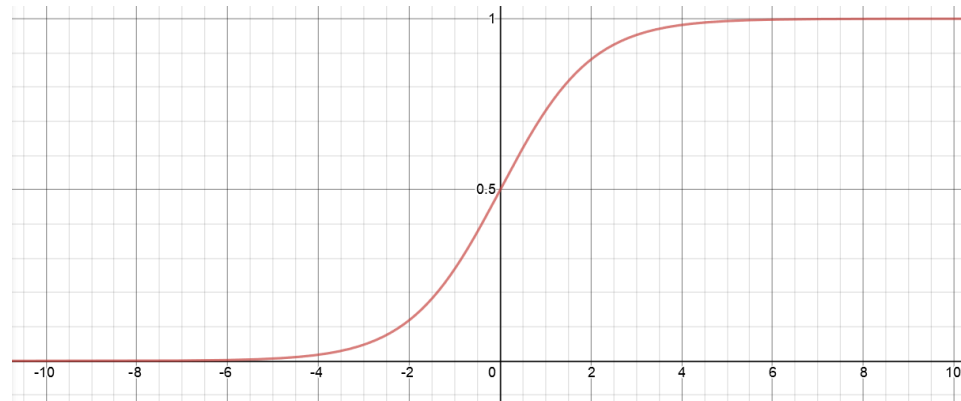
Logistic regression

- Logistic regression uses regression-based techniques for **classification**
- Linear regression function: $w \cdot x$
- Logistic regression function: $g(w \cdot x)$
 - Where g is the **logistic** function $g(z) = \frac{1}{1+e^{-z}}$



Logistic function

- Logistic (or **sigmoid**) function $g(z) = \frac{1}{1+e^{-z}}$
- Nice properties:
 - Smooth and monotonic
 - Bounded range (0,1)
 - Linear at $z=0$: $g'(z) = \frac{z}{(1+e^{-z})^2}$, denominator almost constant

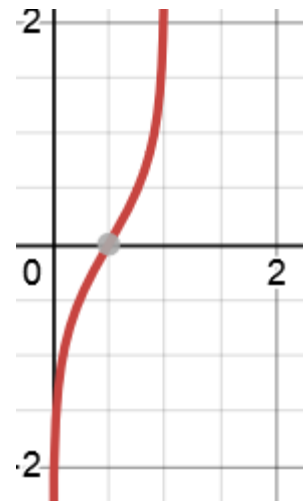


- Why use the logistic function?

- Odds function: $\frac{P(x)}{1-P(x)}$

- Logs-odds function: $\log(P(x)/1-P(x))$

- Logs-odds maps a probability to the infinite real time



- Logistic function is the inverse of the logs-odd function

- So it maps the infinite line to finite probability

Logistic regression

- Want to choose w maximizing the probability of the model assigning to our sample its empirical label:

$$\prod_{x_i: y_i=1} g(w \cdot x_i) \prod_{x_i: y_i=0} [1 - g(w \cdot x_i)]$$

- By taking log, equivalent to maximizing

$$\begin{aligned} \sum_{x_i: y_i=1} \log g(w \cdot x_i) + \sum_{x_i: y_i=0} \log(1 - g(w \cdot x_i)) \\ = \sum_{x_i} \log[(1 - y_i) + (2y_i - 1)g(w \cdot x_i)] \end{aligned}$$